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THE UNIVERSITY OF HONG KONG

Week 2026-W15 to W16 (04/06-19/12)

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APR 2026

Department of Mechanical Engineering (IDT)



Weekly Progress Overview

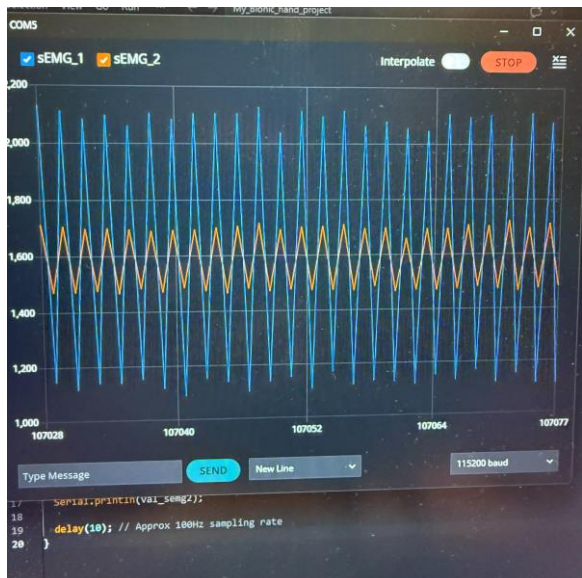
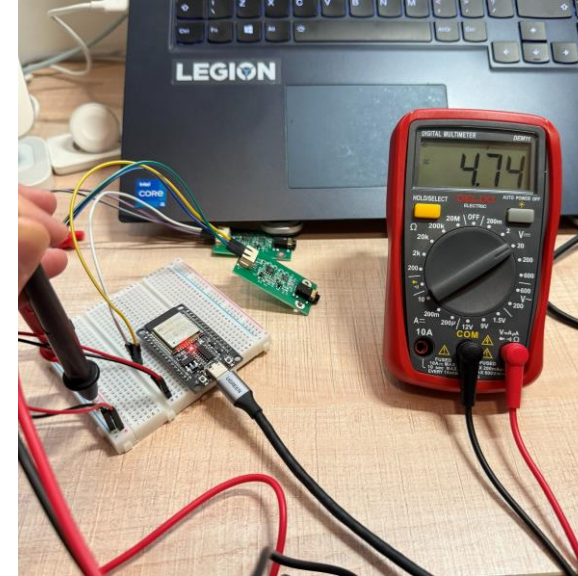
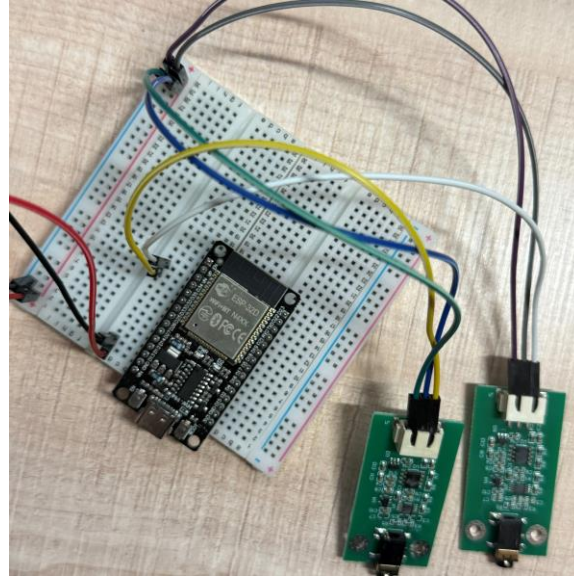
Finished

- Procured necessary sensors and verified hardware compatibility
- Completed hardware assembly for the multi-channel signal collector
- Drafted the initial SVM.py scripty for classification
- Generate gesture labels and data visualisation
- Collected sets of datasets and evaluated process
- Drafted the initial Random_forest.py scripty for classification

Demonstrations



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- Hardware setup completed
- Able to detect muscle signal for both inner and outer side of forearm

Demonstrations

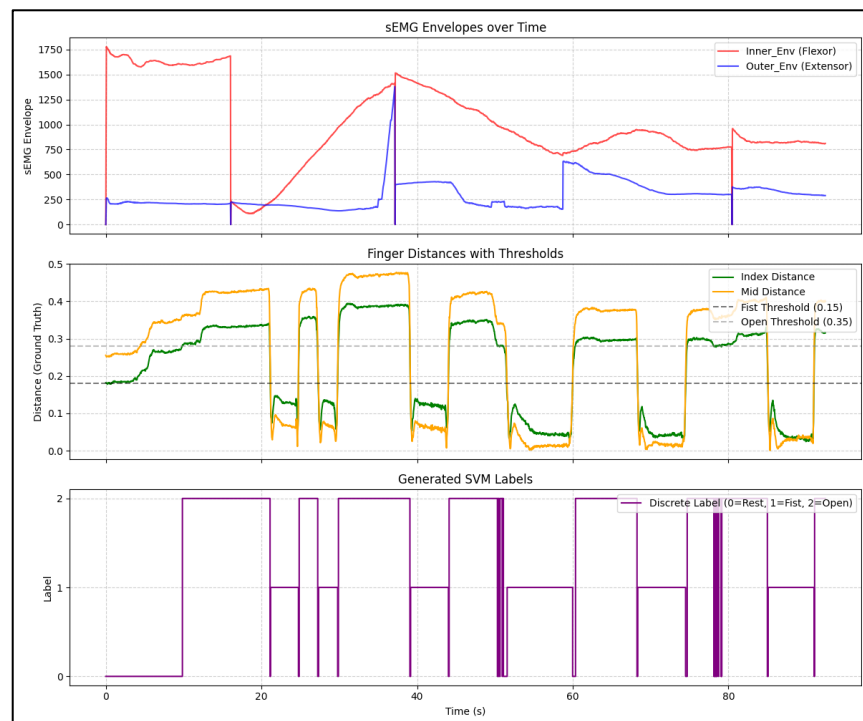


	A	B	C	D	E	F	G	H
1	Timestamp	Inner_Env	Outer_Env	Index_Dist	Mid_Dist	Ring_Dist	Pinky_Dist	
2	1.78E+09	0	0	0	0	0	0	
3	1.78E+09	487.9	582.44	0	0	0	0	
4	1.78E+09	487.2	582.52	0	0	0	0	
5	1.78E+09	487.8	583	0	0	0	0	
6	1.78E+09	487.74	582.96	0	0	0	0	
7	1.78E+09	485.84	583.46	0	0	0	0	
8	1.78E+09	485.54	583.16	0	0	0	0	
9	1.78E+09	485.44	583.82	0	0	0	0	
10	1.78E+09	485.24	583.56	0	0	0	0	
11	1.78E+09	486.02	584.2	0	0	0	0	
12	1.78E+09	485.94	584.06	0	0	0	0	
13	1.78E+09	484.6	584.36	0	0	0	0	
14	1.78E+09	484.4	584.06	0	0	0	0	
15	1.78E+09	482.42	584.1	0	0	0	0	
16	1.78E+09	481.88	584.16	0	0	0	0	
17	1.78E+09	481.1	584.42	0	0	0	0	
18	1.78E+09	481.08	584.42	0	0	0	0	
19	1.78E+09	479.2	584.3	0	0	0	0	
20	1.78E+09	477.04	584.44	0	0	0	0	
21	1.78E+09	476.6	584.42	0	0	0	0	
22	1.78E+09	476.18	584.62	0	0	0	0	
23	1.78E+09	473.42	584.52	0	0	0	0	
24	1.78E+09	472.42	584.52	0	0	0	0	
25	1.78E+09	472.84	584.7	0	0	0	0	
26	1.78E+09	472.08	584.74	0	0	0	0	
27	1.78E+09	471.44	585.26	0	0	0	0	
28	1.78E+09	471.68	585.24	0	0	0	0	

	A	B	C	D	E	F	G	H
777	177554736	643.48	167.44	0.3576369	0.431049	0.4235984	0.4379305	2
778	177554736	647.52	167.3	0.3591368	0.4341266	0.4241939	0.4382366	2
779	177554736	647.06	167.4	0.3589553	0.4333615	0.4231107	0.4377280	2
780	177554736	651.96	167.38	0.3574994	0.4320414	0.4208599	0.436065	2
781	177554736	653.5	167.5	0.3576186	0.4344607	0.4235675	0.4392299	2
782	177554736	658.9	167.18	0.3548451	0.4303529	0.4201278	0.4373920	2
783	177554736	659.82	166.98	0.3535086	0.4311845	0.4200310	0.4359571	2
784	177554736	664.8	166.84	0.3540298	0.4306729	0.4186885	0.4347907	2
785	177554736	666.48	166.84	0.3544211	0.4306393	0.4207183	0.4364067	2
786	177554736	670.42	166.32	0.353978	0.4306815	0.4199731	0.4353550	2
787	177554736	672.16	166.18	0.3548544	0.4308064	0.4189780	0.434477	2
788	177554736	676.7	165.66	0.354203	0.4296363	0.4181375	0.4345119	2
789	177554736	677.54	165.4	0.3541237	0.4305303	0.4192243	0.4348716	2
790	177554736	680.9	164.92	0.3555795	0.4325837	0.4196924	0.4351307	2
791	177554736	681.8	164.68	0.3580921	0.4342858	0.4207586	0.4361980	2
792	177554736	685.58	164.08	0.3569087	0.4339147	0.4205425	0.4362290	2
793	177554736	687.86	164.16	0.3559132	0.4322306	0.4207482	0.4353037	2
794	177554736	692.9	163.8	0.3590656	0.4329232	0.4213041	0.4360540	2
795	177554736	694.8	164.38	0.3570290	0.4322621	0.4206975	0.4357395	2
796	177554736	697.88	163.98	0.3569747	0.4333908	0.4224415	0.4374225	2
797	177554736	698.88	164.1	0.3552344	0.4306632	0.4197935	0.4364655	2
798	177554736	704.66	163.5	0.3546383	0.4307017	0.4204183	0.4375422	2
799	177554736	706.28	163.52	0.3572962	0.4322638	0.4214023	0.4378818	2

- Developed sync_collector.py for real-time data acquisition
- Achieved temporal alignment between serial sEMG data and visual distance measurements
- Achieved labelling of the motion based on data of fingers' distance
- Generate visualised graph based on dataset

Demonstrations



Model accuracy: 0.4888

Confusion Matrix:

```
[[ 23 255]
 [109 325]]
```

Classification Report:

	precision	recall	f1-score	support
Fist (1)	0.17	0.08	0.11	278
Open (2)	0.56	0.75	0.64	434
accuracy			0.49	712
macro avg	0.37	0.42	0.38	712
weighted avg	0.41	0.49	0.43	712

Observations

- Initial segment anomaly
- Non-constant signal consistency
- Rapid state transitions

Improvement plans

- Eliminate initial “unstable data” to improve training set purity
- Changing the raw signal amplitudes into relative intensity to mitigate the impact of signal drift
- Adding window frame to eliminate rapid transitions
- Extend dataset size



Blockers / Questions

- Literature Review: Current theoretical understanding of sEMG control requires deepening; this will be an ongoing background focus.
- Baseline Model Request: Before developing custom optimisations, it is prudent to replicate a mature, open-source sEMG project.
- Discussion point: Could the supervisor recommend a reliable open-source repository as a starting baseline?



Next Week Preview

- Improve dataset for visualisation and for model training
- Conduct a quantitative performance comparison between the SVM and Random Forest models and refine the signal preprocessing pipeline accordingly.